

Deep Learning Image Classification Using TensorFlow/Keras

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GitHub Repo	github.com/DhanavathNavadeep/task-2-Deep_Learning_Image_Classification		

Objective

The objective of this project is to develop an image classification model using Deep Learning techniques. A Convolutional Neural Network (CNN) was trained to classify images into their respective categories with high accuracy. The project demonstrates the complete workflow of image preprocessing, model training, evaluation, and prediction.

Tools and Technologies

Python

TensorFlow / Keras

NumPy

Matplotlib

OpenCV

Scikit-learn

Project Workflow

1. Data Preparation

- Loaded the image dataset efficiently.
- Resized images to a fixed input size suitable for network specifications.
- Normalized pixel values to stabilize gradient descent during backpropagation.
- Split the dataset systematically into training and validation sets.

2. Model Development

A Convolutional Neural Network (CNN) was developed using TensorFlow/Keras. The network architecture consists of sequential convolutional layers for localized feature extraction, pooling layers for spatial dimensionality reduction, fully connected dense layers for high-level abstraction, and a softmax output layer for category probability distribution.

3. Model Training

The model was trained for multiple epochs using the training dataset while validating performance on the validation dataset. Training and validation accuracy and loss metrics were meticulously recorded during each epoch to monitor convergence.

4. Model Evaluation & Saving

The trained model achieved high classification performance with excellent training and validation accuracy. Loss values decreased significantly during training, indicating successful learning. After completion, the pipeline exports the weights as `model.keras` for immediate inference.

Results & Performance

- **Training Accuracy:** Approximately 98%
- **Validation Accuracy:** Approximately 97–99%
- **Loss Metrics:** Exceptionally low training and validation loss
- **Convergence:** Successful convergence of the model without notable overfitting

The generated training curves demonstrate that the model learned effectively, establishing proper boundaries and generalizing robustly to the validation dataset.

Project Output Files

File Name	Description
<code>model.keras</code>	Trained Deep Learning model binaries saved for reuse.
<code>training_curves.png</code>	Generated plots tracking accuracy and loss across epochs.
<code>inference.py</code>	Script execution engine for classifying new input image predictions.

File Name	Description
<code>train.py</code>	The primary execution script containing the model setup and training loops.
<code>requirements.txt</code>	Manifest containing all explicit Python library dependencies.

Challenges Faced

- **Dataset Preprocessing:** Scaling, uniform reshaping, and optimal normalization of high-dimensional images.
- **Dependency Management:** Balancing explicit TensorFlow version alignment and GPU/CPU configurations.
- **Environment Overhead:** Managing Git repository tracking warnings generated by the extensive internal virtual environment (`venv`) layout.
- **Hyperparameter Tuning:** Optimizing structural parameters and regularization boundaries to circumvent overfitting.

Conclusion

The Deep Learning Image Classification model was successfully developed and trained using TensorFlow/Keras. Achieving an exceptional ~98% training accuracy and 97–99% validation accuracy, it validates excellent generalization capacity. The engineering cycle successfully encompasses the end-to-end Deep Learning pipeline—including data preprocessing, model execution, evaluation graphics, serialization, and inference pipeline construction—fully satisfying the technical specifications of Task 2.